BOARD OF INTERMEDIATE EDUCATION, KARACHI

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

SECTION 'A' (Multiple Choice Questions) (40)

> Note: (Options in the original are separated by star marks, shown here as A–D in printed order. Check marks on some options in the scan are a previous solver's marks and are not transcribed. Mathematical notation is transcribed in plain text as printed. The section header says 40 MCQs but the recovered source scan contains only 29 (i–xxix); transcribed as-is.)

> Note: TIME: 3 Hours — (100 Marks) — MATHEMATICS 2022

Q. SECTION 'A' (Multiple Choice Questions) (40)

Q. 1. Choose the correct answer for each from the given options:

Qi. The general term of the sequence 1-2, 2-3, 3-4, ...is:

(A) n
(B) n + 1
(C) n(n + 1)
(D) n(n + 2)

Qii. A real value function f(x) = x + x3 is a/an:

(A) Even function
(B) Odd function
(C) Composite function
(D) Neither even nor odd function

Qiii. lim (n→∞) (1 + 1/n)^n is equal to:

(A) 1
(B) 1/e
(C) e
(D) -e

Qiv. lim (n→∞) $(x^2 - 4)/(x - 2)$

(A) 0
(B) 1
(C) 2
(D) 4

Qv. Slope of a vertical line is:

(A) 0
(B) 1
(C) -1
(D) ∞

Qvi. If the lines a1x + b1y + c1 = 0 and a2x + b2y + c2 = 0 are perpendicular, then:

(A) a1b2 - a2b2 = 0
(B) a1a2 + b1b2 = 0
(C) a1b1 + a2b2 = 0
(D) a1b1 - a2b2 = 0

Qvii. Slope of the line perpendicular to the line 2x - y + 3 = 0 is:

(A) 2
(B) 1/2
(C) -1/2
(D) -2

Qviii. The equation x(1/2) + y(√3/2) = 2 is in perpendicular form, then the value of sin α is:

(A) √3/2
(B) 1/2
(C) 1/√3
(D) 2/√3

Qix. The point of concurrency of the internal bisectors of angles of a triangle is called:

(A) orthocenter
(B) incentre
(C) centroid
(D) circumcenter

Qx. If A, B and C are collinear points then area of triangle ABC is:

(A) Zero
(B) Positive
(C) Negative
(D) infinite

Qxi. Every linear equation represents a/an:

(A) Straight line
(B) Circle
(C) Parabola
(D) Ellipse

Qxii. The inclination of a line is 135o, then slope is:

(A) 1
(B) √3
(C) 0
(D) -1

Qxiii. Two lines represented by $ax^2 + 2hxy + by^2 = 0$ will be orthogonal if:

(A) a + b = 0
(B) a = b
(C) a + b = 1
(D) ab = 0

Qxiv. The centroid of a triangle divides the median internally in the ratio of:

(A) 1 : 2
(B) 3 : 2
(C) 2 : 1
(D) 2 : 2

Qxv. An equation including dy/dx is called:

(A) Polynomial Equation
(B) Differential Equation
(C) Exponential Equation
(D) Logarithmic Equation

Qxvi. If $y(x) = \tan^{-1} 2x$ then $f(x) =$

(A) $1/(1 + x^2)$ (B) $1/(4 + x^2)$

(C) $2/(1 + 4x^2)$ (D) $2/(4 + x^2)$

Qxvii. The slope of tangent to the curve y = x2 + 2x + 3 at (1, 2) is:

(A) 0 (B) 2

(C) 4 (D) 6

Qxviii. If $S = 9\sqrt{t}$ is the relation between distance and time, then speed at t = 1 is:

(A) 9/2 (B) 2/9

(C) 1/18 (D) -2/9

Qxix. If $y = \sin^{-1} 2x$, then its derivative is:

(A) $-2/\sqrt{(1 - 4x^2)}$ (B) $2/\sqrt{(1 - 4x^2)}$

(C) $-2/\sqrt{(1 + 4x^2)}$ (D) $2/\sqrt{(1 + 4x^2)}$

Qxx. If $y = x^3 + 4$, then $d^2y/dx^2 =$:

(A) $3x^2$ (B) 6

(C) 3x (D) 6x

Qxxi. The necessary condition for function f(x) to have extreme value is:

(A) f'(x) = 0 (B) f(x) = 0

(C) f''(x) = 1 (D) f''(x) = 0

Qxxii. Anti-derivative of zero is:

(A) Zero (B) Constant

(C) x (D) 1

Qxxiii. $\int dx/(2x + 3) =$

(A) 2ln(2x + 3) + C (B) 1/2 ln(2x + 3) + C

(C) ln(2x + 3) + C (D) 3ln(2x + 3) + C

Qxxiv. $\int \ln e\hat{}x\, dx =$

(A) ex + C (B) ln ex + C

(C) $x^2/2 + C$ (D) ln x + C

Qxxv. $\int x\hat{}p\, dx =$ when $p \neq -1$:

(A) $x\hat{}(p+1)/(p+1) + C$ (B) $px\hat{}(p-1) + C$

(C) $x\hat{}(p-1)/(p-1) + C$ (D) $px\hat{}(p+1) + C$

Qxxvi. $\int \tan x \sec^2 x \, dx$:

(A) $1/2 \tan^2 x + C$

(B) $1/3 \tan^2 x + C$

(C) $1/2 \sec^2 x + C$

(D) $1/3 \sec^2 x + C$

Qxxvii. $\int e^{(x^2)} 2x \, dx =$

(A) $e^{(x^2)} + C$

(B) $e^{(x^2)}/2 + C$

(C) $2e^{(x^2)} + C$

(D) $e^{(x^2)}/(2x) + C$

Qxxviii. Formula for area above the x - axis, under the curve is:

(A) $\int_a^b y \, dx$

(B) $\int_a^b x \, dx$

(C) $\int_a^b y \, dy$

(D) $\int_a^b x \, dy$

Qxxix. The centre of the circle $x^2 + y^2 + 4x + 2y - 3 = 0$ is

(A) (2, 1)

(B) (-2, -1)

(C) (2, -1)

(D) (1, 2)

SECTION 'B' + SECTION 'C' (Subjective)

Q. SECTION 'B' (Short-Answer Questions) (40)

> Note: NOTE: Attempt any Ten part questions in all. All questions carry equal marks. (i.e. 4 marks of each part).

Q. SECTION "C" (Detailed Answer Questions) (20)

> Note: NOTE: Attempt any TWO question from this section. All questions carry equal marks.